

Adarsh Pradeep

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LinkedIn | GitHub | Portfolio

Skills

Programming: Python, C++, C

Computer Vision: OpenCV, Image Processing, Geometry & Perspective, Feature Extraction

Machine Learning: PyTorch, Scikit-learn, Model Optimization, Multimodal Models (Vision + Text), ASR Pipelines

AI Concepts: CNNs, Embeddings, Similarity Search, Quantization, Pruning, CLIP, FAISS

Systems & Performance: Linux, Shell, WASM basics, Performance Debugging

Web/App: Flutter, Streamlit, HTML, CSS, JavaScript

Tools & Platforms: Git, Firebase, Azure, VS Code

Database: SQL

Experience

Virtusa

May 2026 – Jul 2026

Generative AI Engineer Intern

Chennai, Tamil Nadu

- Developed and optimized LLM-powered workflows using modern Generative AI frameworks and APIs.
- Built AI-driven automation pipelines integrating retrieval, prompt engineering, and backend orchestration.
- Improved inference efficiency and response quality through structured prompting and workflow optimization.

Travelitin Services

Jun 2025 – Present

App Developer (Flutter / Backend)

Remote

- Developed real-time routing and dynamic itinerary generation by optimizing backend logic, improving API response time by 32%.
- Implemented Firestore sync pipelines and an event-driven alert system, ensuring low-latency data updates across all clients.
- Redesigned UI/UX flows, measurably reducing navigation overhead and user drop-off rates.

Liten Technologies

Aug 2025 – Present

App Developer

Remote

- Built scalable Flutter modules integrating secure authentication and optimized REST API data flow, reducing load latency.
- Designed reusable UI components with efficient state management, improving rendering performance across the app.

Projects

Multimodal Rug Search & Recommendation System — Python, CLIP, FAISS, Streamlit Mar – May 2026

github.com/Adxrsh-17/room2rug-multimodal

- Built a **two-pipeline AI search system** over a 68-product catalog supporting **free-text queries** and **room-photo-based rug recommendations** using CLIP and FAISS vector indexing.
- Engineered a **hybrid scoring engine** combining regex-based query parsing, SentenceTransformers semantic similarity, and **multimodal fusion scoring** (0.6 image + 0.4 text), improving retrieval precision across both pipelines.
- Validated system reliability with a **46-test end-to-end suite** covering query parsing, CLIP embeddings, FAISS indexing, and fusion scoring — **all passing**.

Lightweight ASR Architectures — PyTorch, LibriSpeech Jun – Oct 2025

github.com/Adxrsh-17/Lightweight-ASR-Architectures

- Built a compact **Conformer-CTC ASR model** optimized for **edge and low-compute** deployment scenarios.
- Achieved **3.95× model compression** and **1.37× faster inference** by applying **quantization, pruning, SVD decomposition**, and weight clustering.
- Constructed a full **preprocessing, training, and evaluation pipeline** on LibriSpeech train-clean-100, enabling reproducible benchmarking.

Slice-Aware 5G Network Traffic Forecasting — Python, PyTorch, Apache Spark, Docker Nov 2025 – Mar 2026

github.com/Adxrsh-17/5g-var-gru-tft-hybrid

- Built a **decoupled 3-phase Big Data pipeline** using **Apache Spark + HDFS** to ingest gigabit-scale PCAP traces and engineer a **44-dimensional KPI feature space** for slice-aware 5G traffic forecasting.
- Designed a **VAR-BiGRU-TFT hybrid architecture** combining linear cross-KPI correlation, bidirectional temporal modeling, and dynamic feature attention, achieving **R^2 of 0.89 (eMBB), 0.85 (URLLC), 0.79 (mMTC)**.

→ Resolved chronological regime-shift failures via a **10-bin Stratified Shuffle Split**, recovering eMBB validation R^2 from **−0.81 to 0.89**, and deployed models via **ONNX** on Spark Structured Streaming for millisecond-latency inference.

Publications

- **Synergizing Wavelet Scattering Transform and Deep Learning for EEG-Based Depression Classification**
IEEE RECCAP IIT Palakkad [May 22–24, 2026](#)

Certifications

- AI for All: From Basics to GenAI Practice — NVIDIA (2025)
- Supervised ML: Regression and Classification — DeepLearning.AI (2025)
- Foundations: Data, Data Everywhere — Google (2024)

Education

Amrita Vishwa Vidyapeetham, Coimbatore 2023 – 2027
B.Tech (Bachelor of Technology) in Computer Science & Artificial Intelligence CGPA: 8.07/10

Technical Interests

- Systems Programming, Linux, Computer Architecture
- Machine Learning, Deep Learning, Speech Processing
- Mobile App Development, Full-Stack Development